

MODULE 1 – VARIABLE FREQUENCY DRIVE OPERATIONS

Q1.

Apply the concept of V/f control to explain how a VFD maintains constant flux in an induction motor. Illustrate with a neat diagram and discuss its suitability for different load types.

Q2.

Explain how vector control improves torque response compared to scalar control. Apply the concept of decoupled flux and torque control in an industrial motor application.

Q3.

Discuss the operational modes of a VFD (PU mode, External mode, Net mode, Jog mode). Explain how each mode is selected and applied in an industrial setup.

Q4.

Explain the process of auto-tuning (online and offline) in a VFD. Apply this concept to justify its importance in improving motor performance and stability.

Q5.

Describe the braking methods available in VFDs (Dynamic and Regenerative braking). Explain their working principles and industrial applicability.

Q6.

Classify different load patterns (constant torque, variable torque, constant power). Apply suitable VFD control modes for each load and justify your selection.

MODULE 2 – INTERFACING

Q1.

Explain the interfacing of a VFD with PLC using digital and analog I/O. Illustrate how open-loop and closed-loop control are implemented.

Q2.

Describe RS-485 and MODBUS communication protocols used in VFD control. Apply these protocols to explain data exchange between PLC and VFD.

Q3.

Discuss the role of Industrial Ethernet and fieldbus systems in VFD networking. Explain how multiple drives can be integrated in a plant automation system.

Q4.

Explain the implementation of PID control within a VFD. Apply this concept to maintain constant pressure in a blower or pump system.

Q5.

Discuss common VFD faults, warnings, and alarms. Explain systematic troubleshooting methods used in industrial environments.

Q6.

Explain brake sequence functions and advanced control features available in modern VFDs. Discuss their application in high-inertia load systems.

MODULE 3 – HUMAN MACHINE INTERFACE (HMI)

Q1.

Explain the architecture of an HMI system and its integration with PLC and VFD. Illustrate with a block diagram.

Q2.

Discuss the steps involved in creating an HMI project, including screen design, object configuration, and tag mapping.

Q3.

Explain alarm and event handling in HMI systems. Apply this concept to a VFD-based motor control system.

Q4.

Describe trend monitoring and data logging features in HMI. Explain their importance in predictive maintenance of VFD-driven systems.

Q5.

Discuss recipe management and scripting functions in HMI applications. Explain their industrial relevance with suitable examples.

Q6.

Explain HMI security features such as user authentication and access control. Discuss their importance in industrial automation systems.
